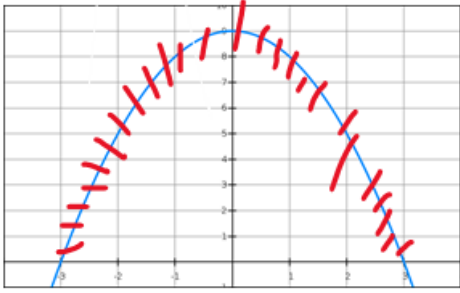
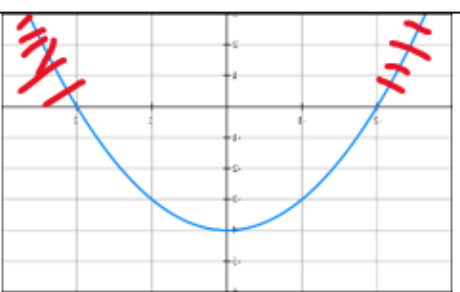
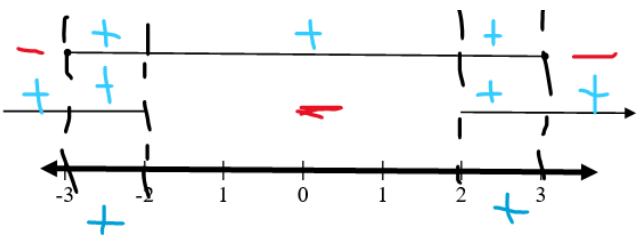
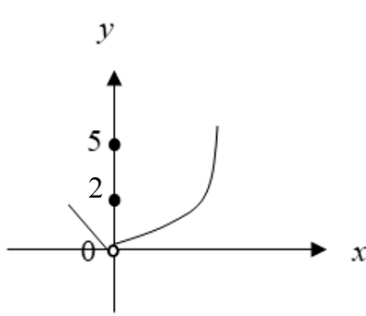


SM015 MATHEMATICS 1
SESSION 2022/2023
ANSWER SCHEME – SET KMPk

No.	Section A
1. (a)	$\lim_{x \rightarrow 1} \frac{-1}{7} \left(\frac{x^3 - 1}{1 - x} \right) = \frac{0}{0} \text{ IDF}$ $\frac{-x^2 - x - 1}{-x + 1} \Bigg _{x^3 - 1}$ $\lim_{x \rightarrow 1} \frac{-1}{7} \left(\frac{(-x^2 - x - 1)(1 - x)}{(1 - x)} \right) = \frac{3}{7}$
(b)	$\lim_{x \rightarrow 3^+} \frac{2 3 - x }{9 - x^2}$ $ 3 - x = \begin{cases} 3 - x, 3 - x \leq 0 \\ -(3 - x), 3 - x > 0 \end{cases}$ $ 3 - x = \begin{cases} 3 - x, x \leq 3 \\ x - 3, x > 3 \end{cases}$ $\lim_{x \rightarrow 3^+} \frac{2(x - 3)}{9 - x^2} = \frac{0}{0} \text{ IDF}$ $\lim_{x \rightarrow 3^+} \frac{-2(3 - x)}{(3 - x)(3 + x)} = -\frac{2}{6} = -\frac{1}{3}$
2.(a)	$5y^2 + 3xy = 45$ $10y \frac{dy}{dx} + 3 \left[y + x \frac{dy}{dx} \right] = 0$ $\frac{dy}{dx} (10y + 3x) = -3y$ $\frac{dy}{dx} = \frac{-3y}{10y + 3x}$ $x = 0, 5y^2 = 45, 5(y - 3)(y + 3) = 0$ $x = 0, y = 3$ $\frac{dy}{dx} = \frac{-3(3)}{10(3) + 3(0)}, \frac{dy}{dx} = -\frac{9}{30} = -\frac{3}{10}$ $x = 0, y = -3$ $\frac{dy}{dx} = \frac{-3(-3)}{10(-3) + 3(0)}, \frac{dy}{dx} = -\frac{9}{30} = -\frac{3}{10}$

(b)	$f(x) = \frac{5}{x^2} + x$ $\lim_{h \rightarrow 0} \frac{\frac{5}{(x+h)^2} + (x+h) - \left(\frac{5}{x^2} + x\right)}{h}$ $\lim_{h \rightarrow 0} \frac{\frac{5x^2 - 5(x+h)^2}{(x+h)^2(x)^2} + h}{h}$ $\lim_{h \rightarrow 0} \frac{h(-10x - 5h)}{h(x+h)^2 x^2} + 1 = \frac{-10x}{x^4} + 1 = \frac{-10}{x^3} + 1$
3.	$y = \frac{x^3}{3} - 2x^2 + 3x + 8$ $y' = 3 \frac{x^2}{3} - 4x + 3 = x^2 - 4x + 3, y'' = 2x - 4$ $y' = 0, (x-1)(x-3) = 0$ $x = 1, x = 3$ $y = \frac{28}{3}, y = 8$ $\left(1, \frac{28}{3}\right), (3, 8) \Rightarrow \text{Stationary Point}$ $x = 1, y''(1) = 2(1) - 4 = -2, \Rightarrow \text{Maximum}$ $\therefore \text{Maximum point} \left(1, \frac{28}{3}\right).$ $x = 3, y''(3) = 2(3) - 4 = 2, \Rightarrow \text{Minimum}$ $\therefore \text{Minimum point}(3, 8).$

No.	Section B
1	$\sqrt{3+4i} = a+bi$ $(\sqrt{3+4i})^2 = (a+bi)^2$ $3+4i = (a^2-b^2) + 2abi$ $a^2-b^2 = 3 \text{ ----} > (1) \qquad ab = 2, \ a^2b^2 = 4 \text{ ----} > (2)$ $b^2(3+b^2) = 4$ $b^4 + 3b^2 - 4 = 0$ $(b^2+4)(b^2-1) = 0$ $b^2 \neq -4, \ b^2 = 1$ $\therefore b = \pm 1, \ a = \pm 2$ $\sqrt{3+4i} = \pm(2+i)$
2	(a) $\frac{1}{x^2-4} - \frac{1}{5} > 0$ $\frac{5-(x^2-4)}{x^2-4} > 0$ $\frac{9-x^2}{x^2-4} > 0$ let $9-x^2 > 0$ $\therefore -3 < x < 3$ and $x^2-4 > 0$ $\therefore x < -2 \text{ or } x > 2$    $\{x: -3 < x < -2 \text{ or } 2 < x < 3\}$ (b) $\frac{7}{7^x} + \frac{7^x}{7} = 2, \ u = 7^x$ $\frac{7}{u} + \frac{u}{7} = 2,$ $\frac{u^2 + 49}{7u} = 2$ $u^2 - 14u + 49 = 0$ $(u-7)^2 = 0$ $7^x = 7^1, \therefore x = 1 \quad \{x: x = 1\}$

3	Let $ax + b = y$ $x = \frac{y-b}{a}$ $f(y) = \frac{2\left(\frac{y-b}{a}\right) + 4}{4\left(\frac{y-b}{a}\right) + 3}$ $f(y) = \frac{2(y-b) + 4a}{4(y-b) + 3a}$ $f(y) = \frac{2y + 4a - 2b}{4y + 3a - 4b}$ $f(x) = \frac{2x + 4a - 2b}{4x + 3a - 4b}$	$\frac{2x + 4a - 2b}{4x + 3a - 4b} = \frac{2(x+3)}{2(2x+1)}$ $4a - 2b = 6, 2a - b = 3 \text{ ----} > (1)$ $3a - 4b = 2, \text{ subs, } b = 2a - 3$ $3a - 4(2a - 3) = 2$ $-5a = -10, a = 2$ $b = 2(2) - 3, b = 1$
4a)	Let $f(x_1) = f(x_2) \therefore 1 - \sqrt{2 - x_1} = 1 - \sqrt{2 - x_2}$ $\sqrt{2 - x_1} = \sqrt{2 - x_2}$ $2 - x_1 = 2 - x_2$ $x_1 = x_2 \therefore f(x) \text{ is one to one and } f(x) \text{ has an inverse.}$ Domain of $f(x) = (-\infty, 2]$ and Range of $f(x)$ is $(-\infty, 1]$	
4 (b)(i)	$[-2, \infty)$	
4 (b)(ii)	 $R_f = (0, \infty)$	
5 (a)	$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (x - 4)$ $= 2 - 4$ $= -2$ $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (2x + B)$ $= 2(2) + B$ $= 4 + B$	
5 (b)	$f(x)$ is continuous at $x = 2$, $f(2) = A$ $\lim_{x \rightarrow 2^-} f(x) = f(2)$ $-2 = A$ $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$ $-2 = 4 + B$ $B = -6$ $\therefore A = -2, B = -6$	

5 (c)	$\lim_{x \rightarrow 5} f(x)$ does not exist $\lim_{x \rightarrow 5^-} f(x) \neq \lim_{x \rightarrow 5^+} f(x)$ $\lim_{x \rightarrow 5^-} (2x - 6) \neq \lim_{x \rightarrow 5^+} C$ $2(5) - 6 \neq C$ $C \neq 4$ $f(x)$ is not continuous at $x = 5$ when $C \neq 4$.
6 (a)	$e^{-2y} + 2(0)y + \ln(1 - 2(0)) = 2$ $e^{-2y} = 2$ $-2y \ln e = \ln 2$ $y = -\frac{1}{2} \ln 2$
6 (b)	$e^{-2y} + 2xy + \ln(1 - 2x) = 2$ $-2 \frac{dy}{dx} e^{-2y} + 2x \left(\frac{dy}{dx} \right) + 2y + \frac{-2}{1 - 2x} = 0$ $-2 \frac{dy}{dx} \left(-2 \frac{dy}{dx} \right) e^{-2y} + e^{-2y} \left(-2 \frac{d^2 y}{dx^2} \right) + 2x \left(\frac{d^2 y}{dx^2} \right) + 2 \left(\frac{dy}{dx} \right) + 2 \frac{dy}{dx} - \frac{4}{(1 - 2x)^2} = 0$ $-\frac{dy}{dx} \left(-2 \frac{dy}{dx} \right) e^{-2y} + e^{-2y} \left(-\frac{d^2 y}{dx^2} \right) + x \left(\frac{d^2 y}{dx^2} \right) + \frac{dy}{dx} + \frac{dy}{dx} - \frac{2}{(1 - 2x)^2} = 0$ $2e^{-2y} \left(\frac{dy}{dx} \right)^2 - e^{-2y} \left(\frac{d^2 y}{dx^2} \right) + x \left(\frac{d^2 y}{dx^2} \right) + 2 \frac{dy}{dx} - \frac{2}{(1 - 2x)^2} = 0$ $(x - e^{-2y}) \frac{d^2 y}{dx^2} + 2e^{-2y} \left(\frac{dy}{dx} \right)^2 + 2 \frac{dy}{dx} - \frac{2}{(1 - 2x)^2} = 0$ Shown
6 (c)	$x = 0, y = -\frac{1}{2} \ln 2$ $-2 \frac{dy}{dx} e^{-2 \left(-\frac{1}{2} \ln 2 \right)} + 2(0) \left(\frac{dy}{dx} \right) + 2 \left(-\frac{1}{2} \ln 2 \right) + \frac{-2}{1 - 2(0)} = 0$ $-2 \frac{dy}{dx} e^{\ln 2} - \ln 2 - 2 = 0$ $-2(2) \frac{dy}{dx} - \ln 2 - 2 = 0$ $\frac{dy}{dx} = -\frac{1}{4} (2 + \ln 2)$ $\frac{dy}{dx} = -0.67$

	$(x - e^{-2y}) \frac{d^2 y}{dx^2} + 2e^{-2y} \left(\frac{dy}{dx} \right)^2 + 2 \frac{dy}{dx} - \frac{2}{(1-2x)^2} = 0$ $\left(0 - e^{-2\left(-\frac{1}{2}\ln 2\right)} \right) \frac{d^2 y}{dx^2} + 2e^{-2\left(-\frac{1}{2}\ln 2\right)} \left(-\frac{1}{4}(2 + \ln 2) \right)^2 + 2 \left(-\frac{1}{4}(2 + \ln 2) \right) - \frac{2}{(1-2(0))^2} = 0$ $(0 - e^{\ln 2}) \frac{d^2 y}{dx^2} + 2e^{\ln 2} \left(-\frac{1}{4}(2 + \ln 2) \right)^2 - \frac{1}{2}(2 + \ln 2) - 2 = 0$ $-2 \frac{d^2 y}{dx^2} + 4 \left(-\frac{1}{4}(2 + \ln 2) \right)^2 - \frac{1}{2}(2 + \ln 2) - 2 = 0$ $-2 \frac{d^2 y}{dx^2} + 4 \left(\frac{1}{16}(2 + \ln 2)^2 \right) - \frac{1}{2}(2 + \ln 2) - 2 = 0$ $-2 \frac{d^2 y}{dx^2} + \frac{1}{4}(2 + \ln 2)^2 - 1 - \frac{1}{2}\ln 2 - 2 = 0$ $\frac{d^2 y}{dx^2} = \frac{3 + \frac{1}{2}\ln 2 - \frac{1}{4}(2 + \ln 2)^2}{-2}$ $\frac{d^2 y}{dx^2} = -0.77$
7 (a)	$72 = 2x^2 y$ $y = \frac{36}{x^2}$ $A = 4x^2 + 4xy + 2xy$ $= 4x^2 + 6xy$ $= 4x^2 + 6x \left(\frac{36}{x^2} \right)$ $A = 4x^2 + \frac{216}{x}$ shown
7 (b)	$\frac{dA}{dx} = 8x - \frac{216}{x^2}$ $0 = 8x - \frac{216}{x^2}$ $8x = \frac{216}{x^2}$ $x^3 = 27$ $x = 3$ $\frac{d^2 A}{dx^2} = 8 + \frac{432}{x^3}$ At $x = 3$, $\frac{d^2 A}{dx^2} = 8 + \frac{432}{3^3} = 24 > 0$ (minimum area) $\therefore x = 3$ $A = 4(3)^2 + \frac{216}{3} = 108 \text{ cm}^3$